Topology optimized bowsprit for an 18-footer dinghy

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The situation

An 18-footer skiff uses a fixed **bowsprit**, which is a spar projecting forward from the bow to be able to project a large downwind sail area that makes these dinghies so fast. The bowsprit is loaded by rope tension, the lift from the sails and the drag of the sails thus creating a combined axial and transverse load on the spar.

The bowsprit must be as stiff and light as possible: excess weight at the bow hurts boat balance, while insufficient stiffness causes the gennaker luff to sag and lose power.

Figure 1: 18-footer skiff with gennaker set. The bowsprit (pointing right) projects forward from the bow; the blue sail tack is attached at its tip. Photo: MySailing [1].

Initial design and constraints

The bowsprit is modelled as a 2D **plane-strain** rectangular domain as shown below. This is appropriate because the spar cross-section is constant and the dominant loading is in-plane.

Bowsprit installation photo — insert image here

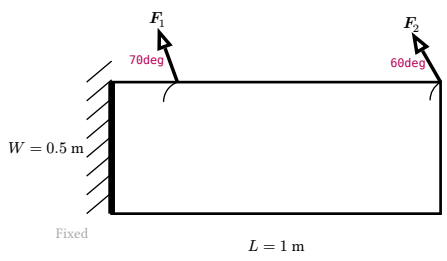

Figure 2: Left: bowsprit installation (photo placeholder). Right: 2D plane-strain optimization domain ($L \times W = 1 \times 0.5$ m). Entire left edge clamped. Surface tractions $\boldsymbol{F}_1$ (70° from horizontal) at 20% of L and $\boldsymbol{F}_2$ (60° from horizontal) at the tip, both directed upper-left.

Boundary conditions and problem data:

- **Clamped** left edge — models the deck/mast-foot fitting.
- **Tip load** at the right-edge centre: resultant of spinnaker tack tension. The load is resolved into a forward component F_x and a downward component F_y based on the tack-line angle α.
- Dimensions: $L = 1$ m (length), $W = 0.2$ m (height/width).
- Volume constraint: $V_{\text{frac}} \leq 33\%$.
- Material: carbon-fibre composite (isotropic approximation), $E = 70$ GPa, $\nu = 0.3$.

Cost function — minimum compliance

Compliance (strain energy) is minimised to maximise stiffness for a given volume of material:

$$J = \int_\Omega \frac{1}{2}\boldsymbol{\sigma} : \boldsymbol{\varepsilon}\, \mathrm{d}\Omega = \frac{1}{2}\boldsymbol{u}^T \boldsymbol{K} \boldsymbol{u}$$

where $\boldsymbol{u}$ is the displacement field, $\boldsymbol{K}$ the global stiffness matrix, $\boldsymbol{\sigma}$ the Cauchy stress and $\boldsymbol{\varepsilon}$ the small-strain tensor.

Solid isotropic material with penalization (SIMP)

Each element carries a design variable $\rho_e \in [\rho_{\min}, 1]$. The effective Young's modulus is penalized as

$$E_e = E_0 \rho_e^p, \quad p = 3$$

to drive intermediate densities toward 0 or 1 (void or solid), producing a manufacturable black–white result. $\rho_{\min} = 10^{-6}$ prevents a singular stiffness matrix.

Filtering and projection

Raw SIMP solutions suffer from **checkerboard** instabilities and mesh-dependent results. A two-step regularization is applied:

Density filter

The design field ρ is convolved with a conic filter of radius $r_{\min}$ to produce a smoothed field $\tilde{\rho}$:

$$\tilde{\rho}_e = \frac{\sum_j w(\boldsymbol{x}_j)\rho_j}{\sum_j w(\boldsymbol{x}_j)}, \quad w(\boldsymbol{x}_j) = \max\big(0, r_{\min} - |\boldsymbol{x}_e - \boldsymbol{x}_j|\big)$$

This removes length-scale violations and guarantees mesh-independence.

Smooth Heaviside projection

$\tilde{\rho}$ is projected to $\overline{\rho}$ with a smoothed step function parameterized by threshold η and sharpness β, producing three designs:

$\eta_D = 0.3$	$\eta_N = 0.5$	$\eta_E = 0.7$
Dilated	*Nominal*	*Eroded*

Table 1: Three robust designs from the Heaviside projection.

If the three designs share the same load-path topology and differ only in member slenderness, the design is considered **robust**: small manufacturing deviations will not change the structural behaviour.

Optimized design

Optimized density field — add image

Figure 3: Nominal optimized bowsprit topology ($\eta_N = 0.5$). Blue = solid, white = void.

The optimizer removes material from the low-stress interior while retaining the outer flanges (high bending stress) and a diagonal web (shear transfer), closely resembling an I-section or truss — consistent with classical beam theory.

Results

Metric	Full solid	Holy beam	Optimized
Max. disp. (mm)	*x.xx*	*x.xx*	*x.xx*
Strain energy (J)	*x.xxx*	*x.xxx*	*x.xxx*
Volume fraction	1.000	*0.xxx*	0.330

Table 2: Performance comparison of beam variants.

Discussion

The optimized design achieves the volume target of 33 % while significantly reducing compliance compared to the uniform holy beam. The resulting topology places material along the principal stress trajectories: flanges carry bending, and diagonal struts carry shear — a result directly analogous to a Michell truss.

Sharp re-entrant corners appear at the clamped edge; in practice these would be filleted to avoid stress concentrations. The plane-strain assumption neglects out-of-plane bending (twist from asymmetric spinnaker loads), which is a limitation to address with a 3D model in future work.

Conclusion

Topology optimization via MMA with SIMP penalization, density filtering, and robust Heaviside projection yields a lightweight bowsprit design that maintains structural stiffness within a 33 % volume budget. The method is systematic, mesh-independent, and produces results that are interpretable in terms of classical structural mechanics.

[1] MySailing, "18ft Skiffs Spring Championship Race 2." 2024.